



Using Machine Learning to Detect Autism-Related Mutations in Noncoding Regions

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Introduction

Noncoding DNA sequences are the regions of the genome that do not directly encode proteins. Instead, they include regulatory elements which interact with the coding regions to control their expression.

Mutations in these regions can still manifest in disorders such as autism.

Because there is little data on these regions, it is hard to directly predict mutations. Large Language Models (LLMs) like our project GenoSCOPE (Genomic Selection Coefficient Optimization via Probabilistic Estimation) trained on DNA can **overcome this restriction in order to predict mutations in noncoding regions.**

Background

Bayesian analysis is a method of incorporating prior knowledge into probabilities. LLMs are used to calculate an informed prior probability distribution (the “prior”) used to ultimately predict the **constraint** of a gene, or how much it tolerates mutation. Constraint could potentially be used to predict the effect of certain mutations, including autism and other disorders.

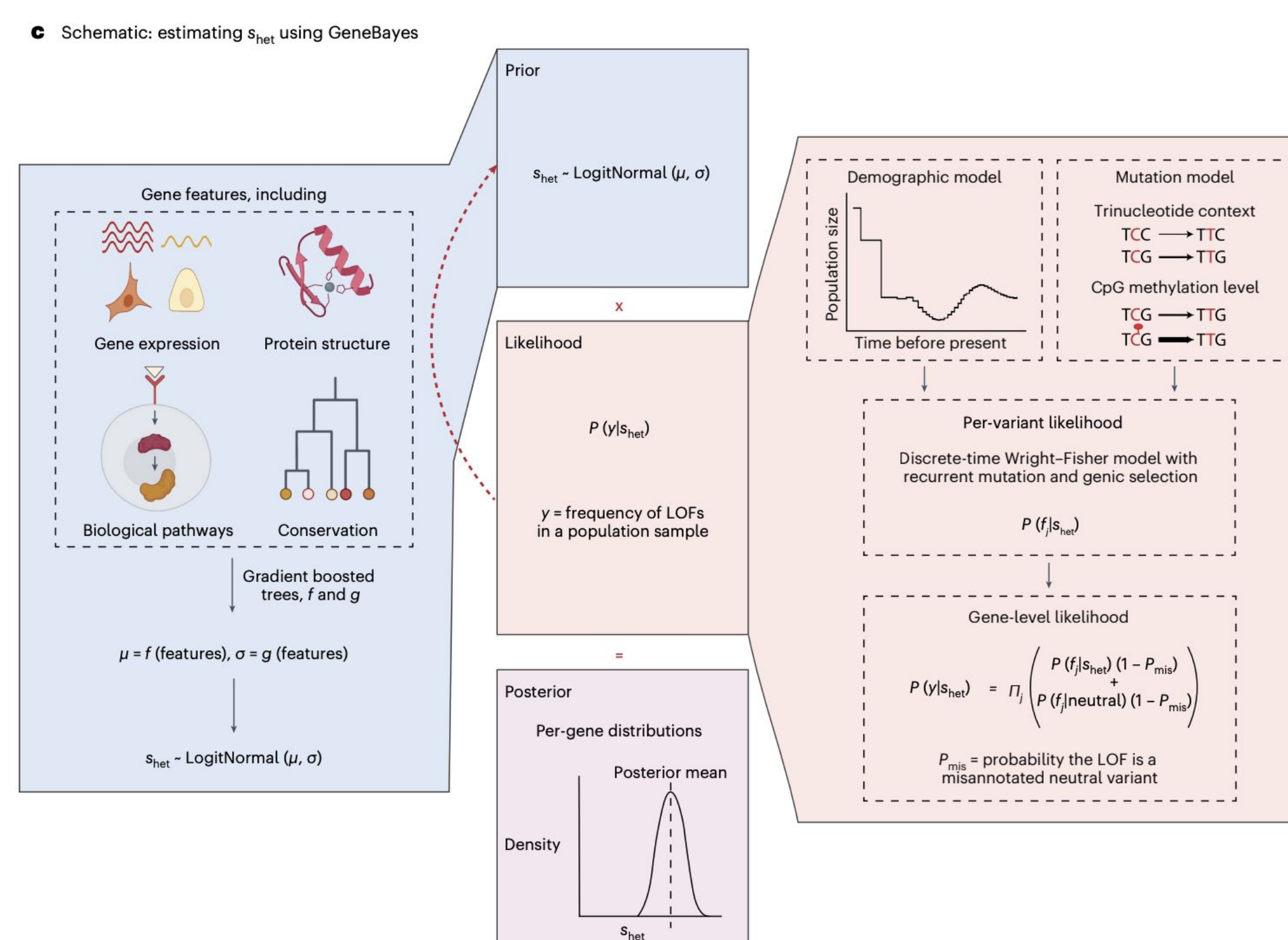


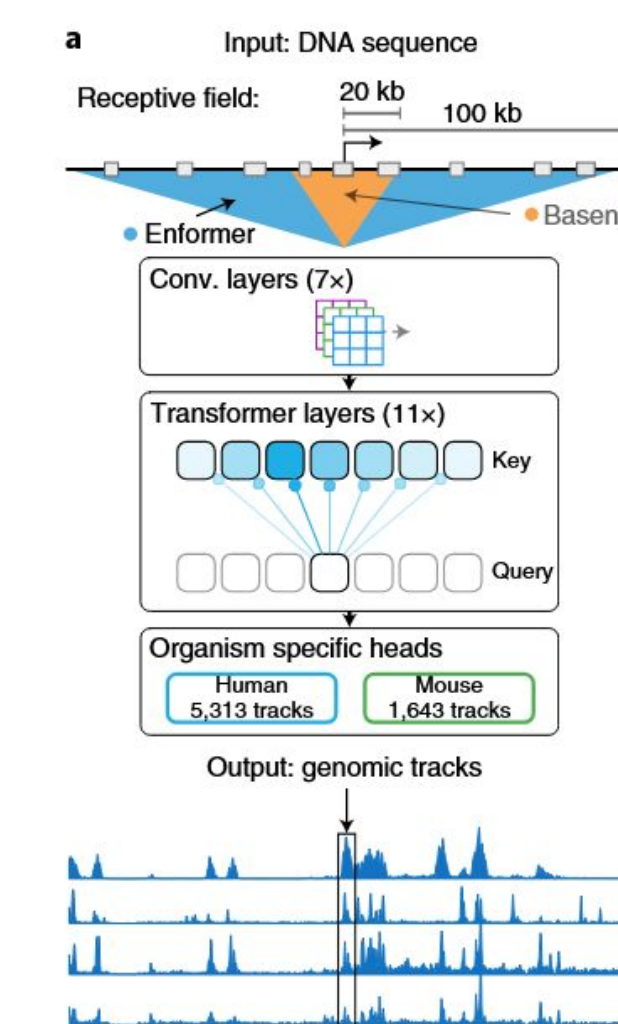
Figure 1: A Bayesian approach to gene expression predictions

For this purpose, the prior used is the number of **observed mutations divided by the number of expected mutations, or o/e**, which is related to constraint: if there are far fewer observed mutations than expected, those mutations would be far more disruptive to the gene, giving it a higher constraint.

Research Question

How can Large Language Models be used to predict mutations in noncoding regions of DNA?

Methodology



GenoSCOPE is trained on a dataset generated by Enformer, which analyzes DNA and outputs epigenomic “tracks”: data describing its features such as gene expression or chromatin openness. Enformer is useful for its large context windows allowing it to properly process noncoding data, which is often spread out. The data is labelled with o/e values for windows of 1000 base pairs.

Figure 2: Enformer architecture and outputted tracks

This model predicts o/e using a regularized linear regression that reduces both the magnitude and number of coefficients, creating a simpler algorithm that is less tailored to specific training data. **Regularization prevents overfitting and bad performance on unseen data**, which previous models struggled with.

Model Agnostic Prediction Interval Estimator (MAPIE) is then used to achieve a probability distribution from the point predictions, which provides the prior for the Bayesian estimation.

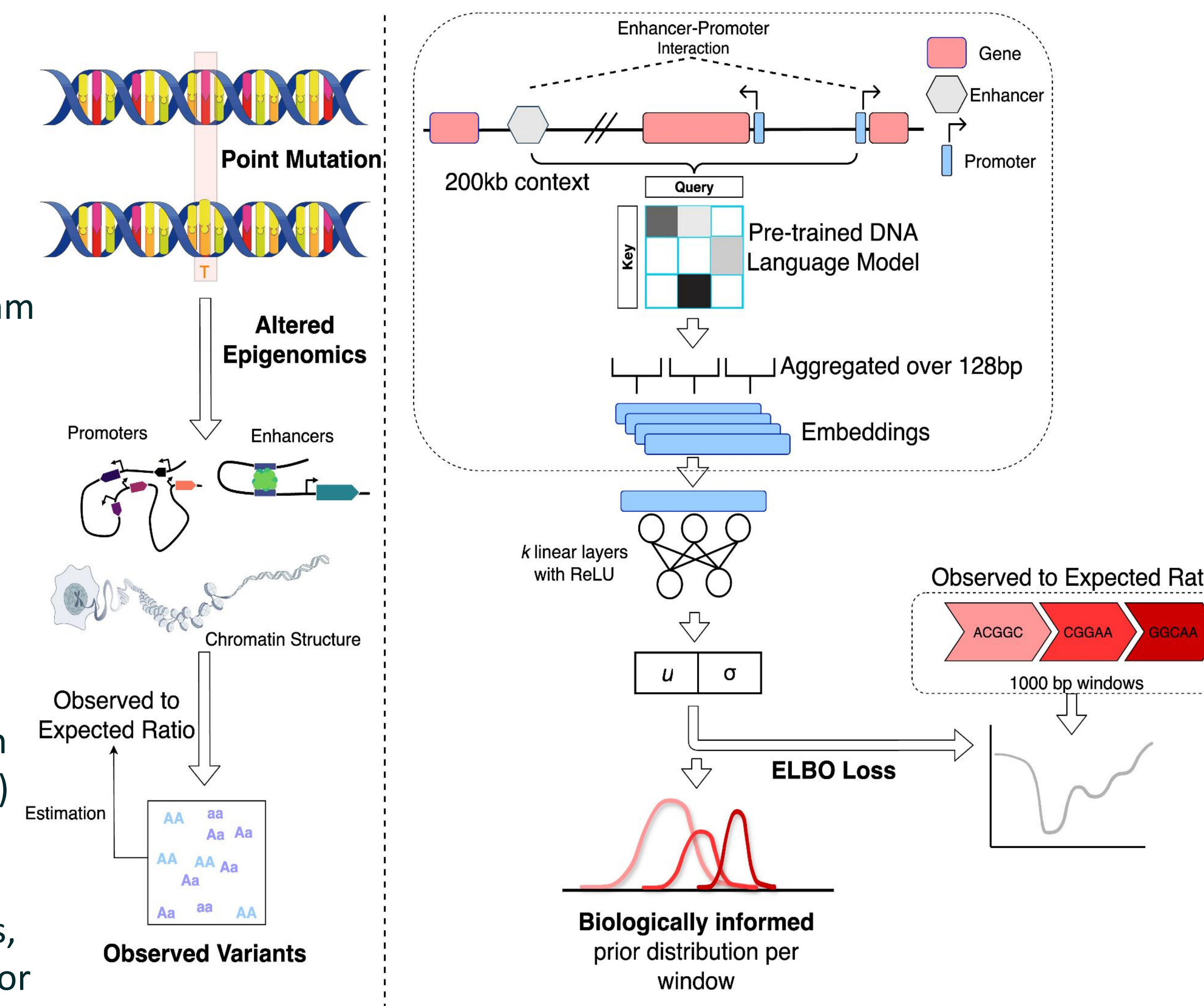


Figure 3: GENOscape processes DNA and predicts constraint, creating an informed prior distribution

References

- [1]: Zeng, T., Spence, J.P., Mostafavi, H. et al. Bayesian estimation of gene constraint from an evolutionary model with gene features. Nat Genet 56, 1632–1643 (2024). <https://doi.org/10.1038/s41588-024-01820-9>
- [2]: Chen, S., Francioli, L. C., Goodrich, J. K. et al. A genomic mutational constraint map using variation in 76,156 human genomes. Nature, 625(7993), 92–100. (2024). <https://doi.org/10.1038/s41586-023-06045-0>

Results/Conclusion

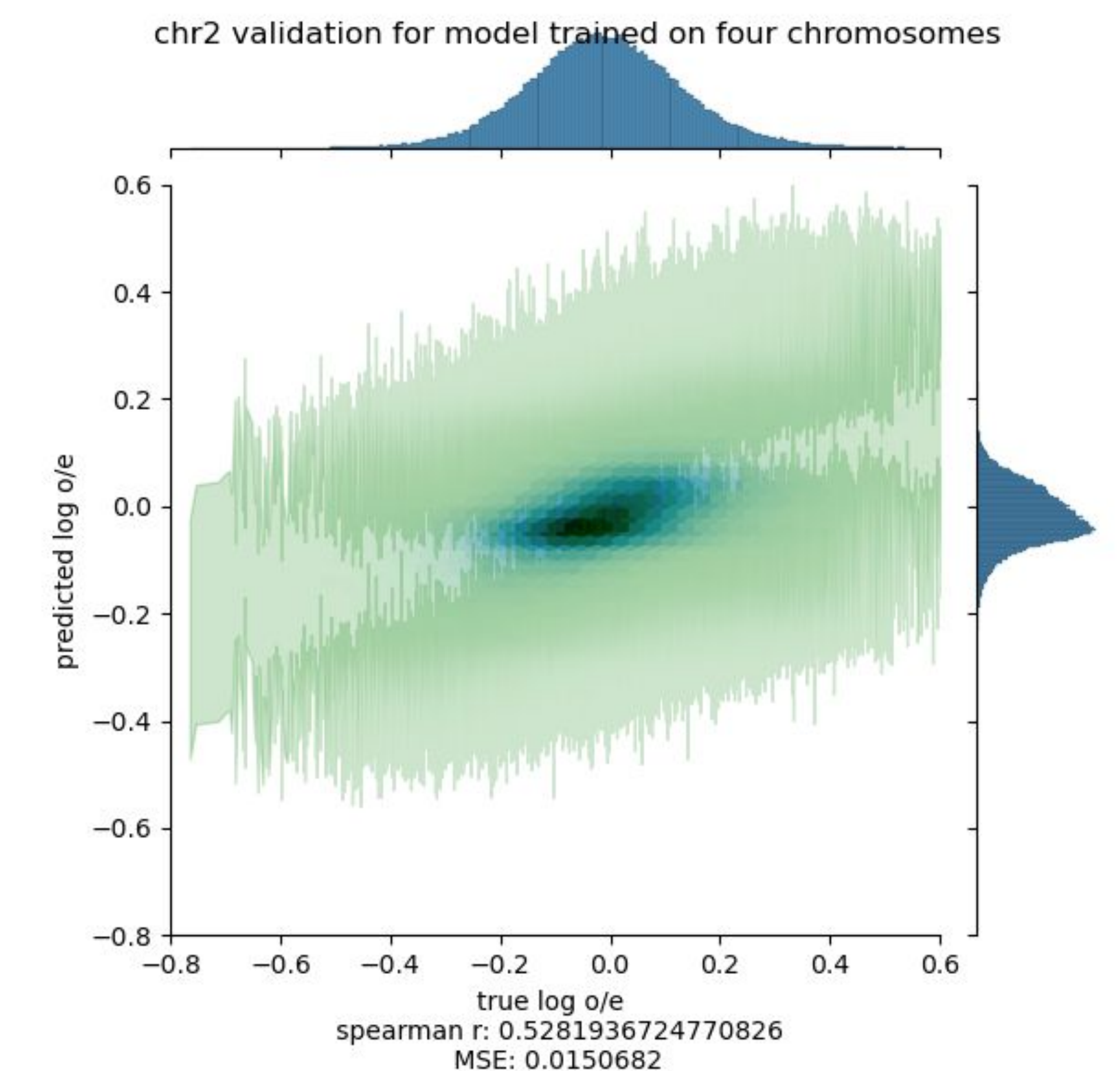


Figure 4: GenoSCOPE's accuracy (predicted/true log o/e)

The blue heatmap shows the concentration of the predicted points, while the transparent green shows the uncertainty distribution for each predicted point.

GenoSCOPE is effective at predicting o/e for unseen data grouped in 1000 base-pair windows without overfitting, working with mutations without extensive data and documentation.

Futures

Currently, window sizes of 1000bp are too large, as regulatory elements range from 100bp to 1000bp. However, as window sizes are reduced, the model becomes more noisy and outliers will affect it more. In order to properly use o/e predictions as a prior to predict gene constraint, we must be able to predict o/e accurately for smaller window sizes.

Acknowledgements

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